

# Bracketing hyperoperations: a rank dichotomy for the Tamari order

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## Abstract

Let  $H_r$  denote the Goodstein hyperoperation hierarchy, so that  $H_1$  is addition,  $H_2$  multiplication,  $H_3$  exponentiation and  $H_4$  tetration. Since  $H_r$  is non-associative for  $r \geq 3$ , the value of a bracketed expression  $H_r(a, \dots, a)$  depends on the bracketing, i.e. on a binary tree. We study how the value varies over the Tamari lattice of such trees.

We prove that a *semi-associative law* runs through the whole hierarchy: for every rank  $r \geq 3$  and all  $x, y \geq 2$ ,

$$H_r(H_r(x, y), z) \leq H_r(x, H_r(y, z)),$$

so a right rotation at any node never decreases the value and evaluation is order-preserving from the Tamari lattice to  $(\mathbb{N}, \leq)$ ; the left comb is a minimiser and the right comb, whose value is  $H_{r+1}(a, n)$ , a maximiser.

Our main result is that the *degeneracy* of this law disappears at rank 4. We classify equality completely. For  $x, y \geq 2$ :

$$r = 3: \quad \text{equality} \iff z = 1 \text{ or } (y, z) = (2, 2); \quad r \geq 4: \quad \text{equality} \iff z = 1.$$

Thus rank 3 carries exactly one non-trivial *equality pattern* — already visible for  $n = 4$  leaves, where the Tamari cover  $((aa)a)a < (aa)(aa)$  has equal endpoints — and from rank 4 on the evaluation is strictly increasing along *every* such cover (Theorem 1.4). (At rank 3 with  $a = 2$  and  $n = 4$  several trees share a value; what is unique is the non-trivial equality pattern  $(y, z) = (2, 2)$  of the local law itself.) The mechanism is identified: the collapsing law  $(x^y)^z = x^{yz}$ , which produces the rank-3 equality, has no analogue at rank 4 or above.

The result is not an asymptotic triviality. The two sides compare a growing *first* argument against a growing *second* one, and small arguments do not blow up with the rank:  $H_r(2, 2) = 4$  for every  $r \geq 2$ . Moreover the hypothesis  $y \geq 2$  is sharp as a uniform hypothesis: for every  $r \geq 3$  and all  $x, z \geq 2$ , taking  $y = 1$  reverses the inequality.

All statements are formalized and machine-checked in Lean 4; the development contains no `sorry`, every theorem is audited with `#print axioms`, and no declaration depends on `Classical.choice`.

## 1 Introduction

Exponentiation is not associative, and it is classical that among all bracketings of  $a^{a^{\dots^a}}$  with  $n$  occurrences of  $a \geq 2$  the right-associated one is the largest; its value is the tetration  $a \uparrow\uparrow n$ . Orders on such bracketings have been discussed before — see Section 8 — but we are not aware of a treatment of the *Tamari* order on them, nor of the behaviour of that order further up the hyperoperation hierarchy.

Two strands of existing work bracket the question. On the one hand, the *associative spectrum* of a binary operation, introduced by Csákány and Waldhauser [1], measures non-associativity by counting how many distinct values the  $C_{n-1}$  bracketings can take; they prove in particular that exponentiation is *Catalan*, i.e. distinct bracketings induce distinct

operations. That line of work counts values. On the other hand, the set of bracketings carries the Tamari lattice [12], generated by right rotations, whose proof-theoretic content was isolated by Zeilberger [13] as a semi-associative law. That line of work orders bracketings. The present paper connects the two: we show that for hyperoperations the Tamari order is *respected* by evaluation, and we locate precisely where the correspondence degenerates.

## Results

Throughout,  $H_r$  is the hyperoperation hierarchy fixed in Section 2, with  $H_1(a, b) = a + b$ ,  $H_2(a, b) = ab$ ,  $H_3(a, b) = a^b$  and  $H_4(a, b) = a \uparrow b$ .

**Theorem 1.1** (Rank 3, complete solution). *Let  $x \geq 1$  and  $y \geq 2$ . Then  $(x^y)^z \leq x^{(y^z)}$  for every  $z \geq 0$ . If moreover  $x \geq 2$ , equality holds if and only if  $z = 1$  or  $(y, z) = (2, 2)$ . The hypothesis  $y \geq 2$  cannot be dropped: for  $y = 1$  and all  $x, z \geq 2$  the inequality fails.*

**Theorem 1.2** (Rank 4). *Let  $x \geq 2$  and  $y \geq 2$ . Then  $(x \uparrow y) \uparrow z \leq x \uparrow (y \uparrow z)$  for every  $z \geq 0$ . If moreover  $z \geq 2$ , the inequality is strict.*

**Proposition 1.3** (Tamari monotonicity). *Let  $a \geq 2$  and  $r \geq 3$ . If  $T \triangleleft U$  is a Tamari cover, i.e.  $U$  arises from  $T$  by a right rotation at some node, then  $\text{ev}_{r,a}(T) \leq \text{ev}_{r,a}(U)$ . Consequently  $\text{ev}_{r,a}$  is order-preserving for the Tamari order. Moreover, for the right comb  $R_n$  with  $n$  leaves,  $\text{ev}_{r,a}(R_n) = H_{r+1}(a, n)$ .*

**Theorem 1.4** (Strict Tamari monotonicity from rank four). *Let  $a \geq 2$  and  $r \geq 4$ . If  $T \triangleleft U$  is a Tamari cover, then*

$$\text{ev}_{r,a}(T) < \text{ev}_{r,a}(U).$$

*Thus for  $r \geq 4$  the evaluation is strictly increasing along every cover of the Tamari lattice, not merely along root rotations.*

**Corollary 1.5** (Rank dichotomy). *For  $a \geq 2$  the evaluation  $\text{ev}_{r,a}$  is order-preserving for the Tamari order at every rank  $r \geq 3$  (Proposition 1.3), and*

- *at rank 3 it is not strict on covers: the cover  $((aa)a)a \triangleleft (aa)(aa)$  carries equal values, both 256, when  $a = 2$ ;*
- *at every rank  $r \geq 4$  it is strict on covers (Theorem 1.4).*

**Theorem 1.6** (All ranks). *For every  $r \geq 3$  and all  $x \geq 2$ ,  $y \geq 2$  and  $z \geq 0$ ,*

$$H_r(H_r(x, y), z) \leq H_r(x, H_r(y, z)).$$

**Theorem 1.7** (Complete equality classification). *Let  $x \geq 2$  and  $y \geq 2$ .*

1. *For every  $r \geq 2$ , equality holds in Theorem 1.6 when  $z = 1$ .*
2. *For  $r = 3$  the only further equality is  $(y, z) = (2, 2)$ .*
3. *For  $r \geq 4$  and  $z \geq 2$  the inequality is strict.*
4. *For every  $r \geq 3$  the inequality is strict when  $z = 0$ , since  $H_r(H_r(x, y), 0) = 1 < x = H_r(x, H_r(y, 0))$ .*

*Items (1), (3) and (4) together give: for  $r \geq 4$  equality holds if and only if  $z = 1$ .*

Theorems 1.6 and 1.7 are proved in Section 6 by inductions on the rank whose engine is the identity-like inequality ( $\text{SUM}_r$ ) below; the case  $r = 3$  is Theorem 1.1 and the case  $r = 4$  can also be obtained directly, as in Section 4.

**Remark 1.8** (The statement is not an asymptotic triviality). Two features are worth isolating. First,  $(\text{HT}_r)$  does not follow from monotonicity alone: on the left the *first* argument grows,  $x \mapsto H_r(x, y) > x$ , while on the right the *second* argument grows,  $z \mapsto H_r(y, z)$ , so the two coordinates compete. Second, high rank does not make everything large: since  $H_r(a, 1) = a$  for  $r \geq 2$ , one gets

$$H_r(2, 2) = H_{r-1}(2, H_r(2, 1)) = H_{r-1}(2, 2) = \cdots = H_2(2, 2) = 4$$

for *every*  $r \geq 2$ . A degenerate corner therefore survives at every rank, and it is exactly this corner that carries the rank-3 equality of Theorem 1.7(2) and that has to be treated separately in the proof of Theorem 1.7(3).

## Formalization

Every theorem stated here is proved in Lean 4 on top of Mathlib. The development defines the hyperoperation by the usual double recursion and *verifies the convention* by proving  $H_1 = +$ ,  $H_2 = \cdot$ ,  $H_3 = ( )^{( )}$ ; this matters because the literature on hyperoperations uses several incompatible indexings and initial values. Section 7 describes the development.

## 2 Preliminaries

**Definition 2.1** (Hyperoperation [7]). Define  $H : \mathbb{N} \times \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$  by

$$H_0(a, b) = b + 1, \quad H_1(a, 0) = a, \quad H_2(a, 0) = 0, \quad H_{r+3}(a, 0) = 1,$$

$$H_{r+1}(a, b + 1) = H_r(a, H_{r+1}(a, b)).$$

**Lemma 2.2.** *For all  $a, b$  we have  $H_1(a, b) = a + b$ ,  $H_2(a, b) = ab$  and  $H_3(a, b) = a^b$ . Moreover  $H_r(a, 1) = a$  for  $r \geq 2$ .*

We write  $a \uparrow\uparrow b := H_4(a, b)$ , so  $a \uparrow\uparrow 0 = 1$  and  $a \uparrow\uparrow (b + 1) = a^{(a \uparrow\uparrow b)}$ .

**Definition 2.3** (Bracketings and evaluation). Let  $\mathcal{T}_n$  denote the set of binary trees with  $n$  leaves. For  $r, a \in \mathbb{N}$  define  $\text{ev}_{r,a} : \mathcal{T}_n \rightarrow \mathbb{N}$  by  $\text{ev}_{r,a}(\text{leaf}) = a$  and  $\text{ev}_{r,a}(T \cdot U) = H_r(\text{ev}_{r,a}(T), \text{ev}_{r,a}(U))$ .

**Definition 2.4** (Right rotation). The *right rotation at the root* sends  $((A \cdot B) \cdot C)$  to  $(A \cdot (B \cdot C))$ . The Tamari order on  $\mathcal{T}_n$  is generated by right rotations performed at arbitrary nodes; that this order is a lattice, with the left comb as least and the right comb as greatest element, is the theorem of Friedman and Tamari [2], whose title already names the relation a *loi demi-associative* [12, 13].

Thus the inequality

$$H_r(H_r(x, y), z) \leq H_r(x, H_r(y, z)) \tag{HT}_r$$

is exactly the assertion that a right rotation, applied with  $x, y, z$  the values of the three subtrees, does not decrease the value.

## 3 Rank three

At rank 3 the collapsing law  $(x^y)^z = x^{yz}$  reduces  $(\text{HT}_r)$  to a comparison of exponents.

**Lemma 3.1.** *If  $y \geq 2$  then  $yz \leq y^z$  for every  $z$ . If moreover  $z \geq 3$  the inequality is strict.*

**Proposition 3.2.** *If  $y \geq 2$  then  $yz = y^z$  if and only if  $z = 1$  or  $(y, z) = (2, 2)$ .*

*Proof.* If  $z = 0$  then  $yz = 0 \neq 1 = y^z$ . If  $z = 1$  both sides are  $y$ . If  $z = 2$  then  $yz = y^z$  reads  $2y = y^2$ , i.e.  $y = 2$  after cancelling  $y > 0$ . If  $z \geq 3$  the inequality is strict by Lemma 3.1. The converse is immediate.  $\square$

*Proof of Theorem 1.1.* Since  $(x^y)^z = x^{yz}$ , the inequality follows from Lemma 3.1 and monotonicity of  $t \mapsto x^t$  for  $x \geq 1$ . For  $x \geq 2$  the map  $t \mapsto x^t$  is injective, so equality holds iff  $yz = y^z$ , and Proposition 3.2 applies. For the last claim,  $(x^1)^z = x^z$  while  $x^{(1^z)} = x$ , and  $x^z > x$  whenever  $x, z \geq 2$ .  $\square$

**Example 3.3** (Failure of strictness). Take  $a = 2$  and  $n = 4$ . The five bracketings evaluate under  $H_3$  to

$$((aa)a)a = 256, \quad (a(aa))a = 256, \quad (aa)(aa) = 256, \quad a((aa)a) = 65536, \quad a(a(aa)) = 65536.$$

In particular the Tamari cover  $((aa)a)a \triangleleft (aa)(aa)$  has equal endpoints, so  $\text{ev}_{3,2}$  is monotone but not strictly monotone; it is also not injective.

## 4 Rank four

At rank 4 no collapsing law is available and a different argument is required. The proof below is by induction on  $z$ ; the base case  $z = 1$  must be treated separately, since the inductive hypothesis at  $z = 0$  is too weak to survive the step.

**Lemma 4.1** (Squaring). *If  $x \geq 2$  and  $m \geq 1$  then  $(x \uparrow\uparrow m)^2 \leq x \uparrow\uparrow (m+1)$ .*

*Proof.* Induction on  $m$ . For  $m = 1$  the claim is  $x^2 \leq x^x$ , which holds as  $x \geq 2$ . For the step put  $t = x \uparrow\uparrow m$ ; then  $(x \uparrow\uparrow (m+1))^2 = (x^t)^2 = x^{2t} \leq x^{(x^t)} = x \uparrow\uparrow (m+2)$ , where  $2t \leq 2^t \leq x^t$  uses  $t \geq 1$  and  $x \geq 2$ .  $\square$

**Remark 4.2.** The direct inequality  $t^2 \leq x^t$  is *false* in general — take  $x = 2, t = 3$  — so the inductive formulation in Lemma 4.1 is essential.

**Lemma 4.3.** *If  $y \geq 2$  and  $m \geq 2$  then  $m+2 \leq y^m$ . If moreover  $m \geq 3$  then  $m+3 \leq y^m$ .*

*Proof of Theorem 1.2.* Write  $y = y' + 1$  with  $y' \geq 1$  and induct on  $z$ . For  $z = 0$  the left side is 1 and the right side is  $x \geq 1$ . For  $z = 1$  both sides equal  $x \uparrow\uparrow y$ . Assume  $z \geq 1$  and set  $m := y \uparrow\uparrow z$ , so  $m \geq y \geq 2$  and  $y' \leq m$ . By Lemma 4.3 we have  $y^m \geq m+2$ , so  $y^m = k+1$  with  $k \geq m+1$ . Writing  $A := x \uparrow\uparrow y = x^{(x \uparrow\uparrow y')}$  we compute

$$\begin{aligned} (x \uparrow\uparrow y) \uparrow\uparrow (z+1) &= A^{(x \uparrow\uparrow y) \uparrow\uparrow z} \leq A^{x \uparrow\uparrow m} = (x^{x \uparrow\uparrow y'})^{x \uparrow\uparrow m} = x^{(x \uparrow\uparrow y')(x \uparrow\uparrow m)} \\ &\leq x^{(x \uparrow\uparrow m)^2} \leq x^{x \uparrow\uparrow (m+1)} \leq x^{x \uparrow\uparrow k} = x \uparrow\uparrow (k+1) = x \uparrow\uparrow (y \uparrow\uparrow (z+1)), \end{aligned}$$

using the inductive hypothesis,  $x \uparrow\uparrow y' \leq x \uparrow\uparrow m$ , Lemma 4.1, and  $m+1 \leq k$ . If  $z \geq 2$  then  $m \geq 4$ , so  $y^m \geq m+3$  by Lemma 4.3 and hence  $m+1 < k$ ; the penultimate step is then strict.  $\square$

**Remark 4.4.** The argument above gives strictness whenever  $y \uparrow\uparrow z \geq 3$ , which covers all  $z \geq 3$  and all  $z \geq 2$  with  $y \geq 3$ . The remaining case is  $y = z = 2$  with  $x \geq 2$  arbitrary, and there strictness holds too:

$$(x \uparrow\uparrow 2) \uparrow\uparrow 2 = (x^x)^{x^x} = x^{x^{x+1}}, \quad x \uparrow\uparrow (2 \uparrow\uparrow 2) = x \uparrow\uparrow 4 = x^{x^{x^x}},$$

and  $x+1 < x^x$  for  $x \geq 2$ . Hence the inequality of Theorem 1.2 is strict for every  $z \geq 2$ ; only the uniform argument degrades at  $y \uparrow\uparrow z = 2$ .

## 5 The dichotomy

*Proof of Corollary 1.5.* Order-preservation at every rank  $r \geq 3$  is Proposition 1.3. Non-strictness at rank 3 is Example 3.3. Strictness at rank 4 is Theorem 1.2; for general  $r \geq 4$  it is Theorem 1.4, which is proved in Section 6 and rests on Theorem 1.7(3). (The present section proves only the rank-4 case, so that the dichotomy is available before the general induction.)  $\square$

*Proof of Proposition 1.3.* Induction on the derivation of the cover. At the root the claim is Theorem 1.6 instantiated at the values of the three subtrees, which are all  $\geq 2$  because  $a \geq 2$ . If the rotation takes place inside the left subtree, one uses monotonicity of  $H_r(\cdot, v)$  in its first argument; inside the right subtree, monotonicity of  $H_r(u, \cdot)$  in its second argument, valid for  $u \geq 2$ . Both monotonicity statements hold for every  $r \geq 1$  and are proved by a double induction on the rank and on the argument. The formula for the right comb is immediate from  $H_{r+1}(a, n+1) = H_r(a, H_{r+1}(a, n))$  together with  $H_{r+1}(a, 1) = a$ .  $\square$

The mechanism is worth stating plainly. At rank 3 the identity  $(x^y)^z = x^{yz}$  turns  $(HT_r)$  into  $yz \leq y^z$ , an inequality which is *tight*: it becomes an equality at  $(y, z) = (2, 2)$ , and this single coincidence is the source of every collapse in Example 3.3. At rank 4 there is no such identity, the comparison is genuinely two-dimensional, and the slack is enormous.

## 6 General rank

We record the general statement and the shape of a proof.

We now prove Theorem 1.6. The engine is the following inequality; write

$$H_{r-1}(H_r(x, a), H_r(x, b)) \leq H_r(x, a + b). \quad (\text{SUM}_r)$$

**Proposition 6.1.** *Let  $r \geq 3$  and  $x \geq 2$ .*

1. *If  $(HT_{r-1})$  holds, then  $(\text{SUM}_r)$  holds for all  $a, b \geq 1$ .*
2. *If  $(\text{SUM}_r)$  holds and  $u + v \leq H_{r-1}(u, v)$  for all  $u, v \geq 2$ , then  $(HT_r)$  holds for all  $y \geq 2$ .*

*Proof.* (1) Induct on  $a$ . For  $a = 1$  both sides are equal:

$$H_{r-1}(H_r(x, 1), H_r(x, b)) = H_{r-1}(x, H_r(x, b)) = H_r(x, b + 1).$$

For the step,  $H_r(x, a + 1) = H_{r-1}(x, H_r(x, a))$ , so by  $(HT_{r-1})$

$$H_{r-1}(H_{r-1}(x, H_r(x, a)), H_r(x, b)) \leq H_{r-1}(x, H_{r-1}(H_r(x, a), H_r(x, b))),$$

and the inductive hypothesis bounds the inner term by  $H_r(x, a + b)$ , giving  $H_{r-1}(x, H_r(x, a + b)) = H_r(x, a + b + 1)$ .

(2) Induct on  $z$ , with  $z = 0, 1$  as in Theorem 1.2. For the step set  $m := H_r(y, z) \geq 2$ ; then  $H_r(H_r(x, y), z + 1) = H_{r-1}(H_r(x, y), H_r(H_r(x, y), z)) \leq H_{r-1}(H_r(x, y), H_r(x, m)) \leq H_r(x, y + m) \leq H_r(x, H_{r-1}(y, m)) = H_r(x, H_r(y, z + 1))$ .  $\square$

*Proof of Theorem 1.6.* Induction on  $r$ . The base case  $r = 3$  is Theorem 1.1. For the step, the inductive hypothesis is exactly  $(HT_{r-1})$ , so Proposition 6.1(1) gives  $(\text{SUM}_r)$ , and Proposition 6.1(2) then gives  $(HT_r)$ , the hypothesis  $u + v \leq H_{r-1}(u, v)$  for  $u, v \geq 2$  being Lemma 6.2.  $\square$

**Lemma 6.2.** *For  $u, v \geq 2$  and  $s \geq 2$  we have  $u + v \leq H_s(u, v)$ .*

*Proof.*  $u + v \leq uv = H_2(u, v)$ , and  $H_2(u, v) \leq H_s(u, v)$  by monotonicity in the rank.  $\square$

## Strictness

**Lemma 6.3.** *For  $u, v \geq 2$  with  $u \geq 3$  or  $v \geq 3$ , and  $s \geq 2$ , we have  $u + v < H_s(u, v)$ . Equality  $u + v = H_s(u, v)$  occurs exactly at  $u = v = 2$ , where both sides are 4.*

*Proof.*  $u + v < uv$  holds iff  $(u - 1)(v - 1) > 1$ , i.e. iff  $u \geq 3$  or  $v \geq 3$ ; and  $uv = H_2(u, v) \leq H_s(u, v)$ . At  $u = v = 2$  we have  $u + v = 4 = H_2(2, 2) = H_s(2, 2)$  by Remark 1.8.  $\square$

*Proof of Theorem 1.7.* (1)  $H_r(H_r(x, y), 1) = H_r(x, y) = H_r(x, H_r(y, 1))$  since  $H_r(a, 1) = a$ . (2) is Theorem 1.1. For (3) we induct on  $r \geq 4$ , the base  $r = 4$  being Theorem 1.2 with Remark 4.4. Let  $r \geq 5$  and write  $z = w + 1$  with  $w \geq 1$ , and  $m := H_r(y, w) \geq 2$ .

If  $y \geq 3$  or  $w \geq 2$ , then  $y \geq 3$  or  $m \geq 4$ , so Lemma 6.3 makes the last step of the proof of Proposition 6.1(2) strict, and strictness propagates because  $H_r(x, \cdot)$  is strictly increasing.

The remaining case is  $y = z = 2$ . Put  $A := H_r(x, 2) = H_{r-1}(x, x) \geq 4$ . Then

$$H_r(H_r(x, 2), 2) = H_{r-1}(A, A), \quad H_r(x, H_r(2, 2)) = H_r(x, 4) = H_{r-1}(x, H_{r-1}(x, A)),$$

using  $H_r(2, 2) = 4$  from Remark 1.8. Since  $A = H_{r-1}(x, x)$ , the required inequality

$$H_{r-1}(H_{r-1}(x, x), A) < H_{r-1}(x, H_{r-1}(x, A))$$

is the inductive hypothesis at rank  $r - 1$  applied to  $(x, x, A)$ , and  $A \geq 2$ .  $\square$

We can now upgrade Proposition 1.3 to its strict form.

*Proof of Theorem 1.4.* Induction on the derivation of the cover, exactly as for Proposition 1.3 but with every inequality replaced by its strict counterpart. At the root the claim is Theorem 1.7(3) instantiated at the values of the three subtrees, which are all  $\geq 2$  because  $a \geq 2$  (so in particular the hypothesis  $z \geq 2$  of that item is met). If the rotation takes place inside the left subtree, one needs

$$2 \leq u < u' \implies H_r(u, v) < H_r(u', v) \quad (v \geq 1),$$

the strict form of monotonicity in the first argument; if it takes place inside the right subtree, one needs

$$1 \leq v < v' \implies H_r(u, v) < H_r(u, v') \quad (u \geq 2, r \geq 2),$$

the strict form of monotonicity in the second argument. Both are proved by the same double induction on the rank and on the argument as their non-strict counterparts, and both apply here because every subtree value is  $\geq 2$ .  $\square$

The auxiliary facts used above — monotonicity of  $H_r(x, \cdot)$  in its second argument, monotonicity in the rank, and  $n + 1 \leq H_r(x, n)$  for  $x \geq 2, r \geq 2, n \geq 1$  — are elementary and are proved in the formalization by a sequence of inductions.

The hypothesis  $y \geq 2$  of Theorem 1.6 is sharp, in the following precise sense: for every  $r \geq 3$  and all  $x, z \geq 2$ , taking  $y = 1$  reverses the inequality. Indeed  $H_r(x, 1) = x$ , so the left side is  $H_r(x, z)$ , whereas  $H_r(1, z) = 1$  makes the right side  $H_r(x, 1) = x$ , and  $H_r(x, z) > x$  because  $z \geq 2$ . The restriction to  $x, z \geq 2$  is necessary and not an artefact: for  $x = 1$  both sides are often equal, for  $z = 1$  they are equal by Theorem 1.7(1), and for  $z = 0$  the left side is 1 and the right side is  $x$ , so the original inequality still holds. For  $x = 0$  the operations degenerate.

## 7 The Lean development

The development is written for Lean 4 (v4.30.0) with Mathlib (v4.30.0) and is available at

<https://github.com/turahigashi/hyper-tamari>

It consists of three files, `HyperTamari/Basic.lean`, `HyperTamari/Tetration.lean` and `HyperTamari/General.lean`, together with the raw lake build output recording the axiom audit (`logs/axiom-audit.txt`).

The correspondence with the numbered statements of this paper is given in Table 1. We distinguish three cases, because they carry different evidential weight:

EXACT a single declaration with the same quantifiers, hypotheses and conclusion as the printed statement;

COLLECTIVE the statement is covered by a combination of declarations (typically because the paper states several clauses at once);

PROSE background material that is *not* formalized here — standard facts about the Tamari lattice that we use as context, or explanatory remarks.

We stress the third row of this classification: the paper does *not* claim that everything it says is machine-checked. What is machine-checked is every inequality and every equality assertion about  $H_r$  and about the evaluation of trees.

The development also contains `pow_lt_pow_right`, `pow_right_injective`, choice-free replacements for the corresponding Mathlib lemmas (see below), and the auxiliary `succ_le_hyper`, `hyper_mono_arg`, `hyper_mono_base` and `hyper_rank_mono`.

**Draft note — not part of the submission.** The artifact has *not* yet been deposited in a long-term archive, so no DOI is quoted above. A version DOI will be minted and inserted here at the moment this paper is submitted, and the archive record will be linked to the submission. Until then the repository URL is the only pointer.

No declaration uses `sorry`, `native_decide` or any private axiom. Every one of the 82 theorems in these three files is audited with `#print axioms`, and the audit is machine-checked to cover them all (declared 82, audited 82, missing 0); `sorryAx` does not occur. The axiom dependencies are `propext` alone (3 theorems) and `propext`, `Quot.sound` (79). In particular

**no declaration depends on `Classical.choice`.**

This is not a foundational claim, and choice-freeness is not what the paper is about: every statement here is an elementary assertion about natural numbers, and nobody would expect it to need choice. We record it because it costs nothing and makes the artifact reusable — a development free of `Classical.choice` can be transported into constructive settings, or used inside a larger development that must itself avoid choice, without reproving anything.

It is worth saying where the choice had been hiding, since it was not in the mathematics. An earlier version of this development did depend on `Classical.choice` in 30 of its declarations, and the dependency was traced to exactly four sources, all of them incidental: the Mathlib lemmas `Nat.pow_lt_pow_right` and `Nat.pow_right_injective` (whose companion `Nat.pow_le_pow_right` is proved with no axioms at all), the `nlinarith` tactic, and — the one that is easiest to miss — `norm_num` applied to a numeric *inequality*, which pulls in choice, while `omega` discharges the same goal without it and `norm_num` applied to an *equation* does not. Replacing these by `omega`, by `decide`, by `Nat.add_le_mul`, and by two short inductions built on the axiom-free `Nat.pow_lt_pow_succ` removed the dependency entirely, with no change to any statement.

| Statement                                               | Lean declaration(s)                                                                                                                                                               | Status     |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Def. 2.1                                                | <code>hyper, hyper_zero, hyper_one_zero, hyper_two_zero, hyper_succ_succ_zero, hyper_succ</code>                                                                                  | EXACT      |
| Lem. 2.2                                                | <code>hyper_one, hyper_two, hyper_three</code>                                                                                                                                    | COLLECTIVE |
| Def. 2.3 (trees, ev)                                    | <code>BTree, BTree.eval</code>                                                                                                                                                    | EXACT      |
| Def. 2.4 ( $\mathcal{T}_n$ , lattice, comb extremality) | — ( <code>BTree</code> is not restricted to a fixed number of leaves; the lattice property and the extremality of the two combs are classical [2, 12] and are used as background) | PROSE      |
| Def. 2.4 (Tamari order)                                 | <code>Rot (cover), Relation.ReflTransGen Rot (the order)</code>                                                                                                                   | EXACT      |
| Thm. 1.1 (inequality, equality iff)                     | <code>ht_rank3, ht_rank3_eq_iff, mul_eq_pow_iff</code>                                                                                                                            | COLLECTIVE |
| Thm. 1.1 (sharpness of $y \geq 2$ )                     | <code>ht_needs_two_le_y</code> — a <i>witness</i> at $x = z = 3$ , not the universal statement                                                                                    | COLLECTIVE |
| Lem. 3.1                                                | <code>mul_le_pow_of_two_le, mul_lt_pow_of_three_le</code>                                                                                                                         | COLLECTIVE |
| Prop. 3.2                                               | <code>mul_eq_pow_iff</code>                                                                                                                                                       | EXACT      |
| Ex. 3.3 (the colliding cover)                           | <code>rank3_rotation_not_strict, rank3_collision_value</code>                                                                                                                     | COLLECTIVE |
| Ex. 3.3 (the five values of $\mathcal{T}_4$ )           | — (only the colliding pair and its value 256 are formalized)                                                                                                                      | PROSE      |
| Lem. 4.1                                                | <code>sq_tet_le</code>                                                                                                                                                            | EXACT      |
| Lem. 4.3                                                | <code>add_two_le_pow, add_three_le_pow</code>                                                                                                                                     | COLLECTIVE |
| Thm. 1.2                                                | <code>tet_ht, tet_ht_strict_of_two_le</code>                                                                                                                                      | COLLECTIVE |
| Rem. 4.4                                                | <code>tet_ht_strict_of_two_le</code>                                                                                                                                              | EXACT      |
| Prop. 1.3 (covers, order)                               | <code>Rot.eval_le, Rot.eval_le_of_reflTransGen</code>                                                                                                                             | EXACT      |
| Prop. 1.3 (right comb)                                  | <code>rightComb, eval_rightComb</code>                                                                                                                                            | EXACT      |
| Prop. 1.3 (combs are extremal)                          | — (classical property of the Tamari lattice, not formalized)                                                                                                                      | PROSE      |
| Thm. 1.4                                                | <code>Rot.eval_lt, Rot.eval_lt_of_transGen, hyper_lt_base</code>                                                                                                                  | EXACT      |
| Cor. 1.5                                                | <code>rank3_rotation_not_strict, Rot.eval_lt</code>                                                                                                                               | COLLECTIVE |
| Lem. 6.2                                                | <code>add_le_hyper</code>                                                                                                                                                         | EXACT      |
| Lem. 6.3                                                | <code>add_lt_hyper</code>                                                                                                                                                         | EXACT      |
| Prop. 6.1                                               | <code>sum_lemma</code>                                                                                                                                                            | EXACT      |
| Thm. 1.6                                                | <code>ht_general</code>                                                                                                                                                           | EXACT      |
| Thm. 1.7                                                | <code>equality_classification</code> (four components), <code>ht_eq_at_one, ht_rank3_eq_iff, ht_strict_general, ht_lt_at_zero</code>                                              | COLLECTIVE |
| Rem. 1.8 ( $H_r(2, 2) = 4$ )                            | <code>hyper_two_two, hyper_one_base</code>                                                                                                                                        | EXACT      |
| Rem. 1.8 (the two coordinates compete)                  | — (explanatory)                                                                                                                                                                   | PROSE      |

Table 1: Correspondence between the numbered statements and the Lean development.

## 8 Related work

Csákány and Waldhauser [1] introduce the associative spectrum and show that exponentiation is Catalan: for pairwise distinct prime leaves, distinct bracketings give distinct values. Their concern is the number of values; the order in which those values arrange themselves is not addressed, and the Tamari lattice appears in their bibliography only as the source of the enumeration of bracketings. Huang and Lehtonen [9] extend the invariant to the *associative-commutative spectrum* and determine it for exponentiation (their Proposition 6.1.5): the associative spectrum of exponentiation is  $C_{n-1}$  and its ac-spectrum is  $n^{n-1}$ . Their question is again how many distinct term operations the bracketings produce; ours is how the resulting values are *arranged*, and neither the Tamari order nor any monotonicity statement occurs in that line of work. Huang, Mickey and Xu [10] compute the associative spectrum of a further operation. Tamari [12] enumerated the bracketings, and Friedman and Tamari [2] proved that the right-rotation order on them is a finite lattice — their title already calls the relation a *loi demi-associative*, which is precisely the law we show to run through the whole hyperoperation hierarchy. Zeilberger [13] gives a sequent calculus for that law. Gryszka [8] studies the comparison of power towers by normalizing them, which concerns right-associated towers rather than general bracketings.

### The rank-three case is not new

It is important to say clearly what part of Proposition 1.3 is already known. In a MathOverflow question of January 2018 [3], Spivak treats exponentiation on the integers  $\geq 2$  as an ordered magma and orders bracketings by the *evaluation order*:  $E \leq F$  when the inequality holds *for every* assignment of values to the leaves. He asks how this order relates to the height (Stanley) order on Dyck paths. In the answers, Chow proves that a Stanley cover does decrease the value, the computation resting on

$$B^{CD} = (B^C)^D \geq (BC)^D \geq BC^D \quad (B, C \geq 2, D \geq 1),$$

and Rubey reduces the same covering relation to the inequality  $L_2^{(L_1^{xR})} \geq (L_2^{L_1})^{(xR)}$ , which is exactly (HT<sub>r</sub>) at rank 3 in a variable-leaf form.

Since the evaluation order quantifies over *all* leaf assignments it is stronger than the single-label evaluation  $\text{ev}_{3,a}$  used here, and since the Stanley lattice is an extension of the Tamari lattice, comparability in the Tamari order implies comparability in the Stanley order. Consequently **the rank-3 case of Proposition 1.3 is already contained, in a stronger form, in that discussion**, and we do not claim it. What that discussion does not contain is any rank other than 3, any analysis of when equality holds, or the strictness phenomenon; those are the subject of this paper.

There is also a body of peer-reviewed work on orders attached to iterated exponentials, and it is worth saying precisely how it relates to the present question, because the titles suggest a closer overlap than there is. Brunson [4] fixes  $e \leq x_1 \leq \dots \leq x_n$  and, *for each permutation* of the  $x_i$ , forms the tower  $T(y, z, \dots, w) = y^{z^{\dots^w}}$ , in which “association is always to the upper right”; the resulting partial order lives on the symmetric group  $S_n$ . Stembridge [11] and Griggs–Wachs [6] compare that order with the Bruhat order on  $S_n$ , the latter exhibiting a counterexample at  $n = 5$  to their coincidence. Griggs [5] studies the poset of towers over the two-letter alphabet  $\{a, b\}^n$ , the tower again being “evaluated in the conventional way, from top down”.

In every one of these the *bracketing is fixed* — it is the right comb — and what varies is the assignment of entries to its leaves. Here the situation is exactly the reverse: the entries are fixed, all equal to  $a$ , and what varies is the bracketing. The two families of questions are

therefore orthogonal, and none of these papers contains the inequality studied here. (None of [4, 5, 6] contains an occurrence of the words *bracketing*, *Tamari*, *binary tree*, *Catalan*, *rotation* or *Dyck*.)

## What we do claim

Not finding a result is not a proof that it is new, so we state the claim in pieces. The rank-3 monotonicity is known (above). For the following we have not found a prior occurrence: the local semi-associative law ( $\text{HT}_r$ ) for *every* rank  $r \geq 3$  of the Goodstein hierarchy; the complete classification of its equality cases, with the isolated rank-3 exception  $(y, z) = (2, 2)$ ; the resulting dichotomy, strict on every Tamari cover from rank 4 on; and the machine-checked formalization.

## Use of AI assistance

In accordance with current disclosure practice, the author states the following.

Artificial-intelligence assistants were used substantially in the preparation of this work. The question studied here was formulated by the author. Large language models were then used to search for the shape of the proofs, to write the entire Lean 4 development, to draft this text, and to check the literature; further language models were used as adversarial readers, and their criticism led to the present form of the main theorem, in particular to the classification of the equality cases in Theorem 1.7. All mathematical decisions, the choice of what to claim, and the final text are the author’s responsibility.

The reader is asked to note what this does and does not mean, and we state it as narrowly as we can. Every inequality and every equality assertion about  $H_r$  and about the evaluation of trees that appears in Sections 3–6 has been formalised and checked by the Lean 4 kernel; Table 1 says, statement by statement, which declaration carries it, and marks the places where the paper relies instead on classical background about the Tamari lattice or on explanatory prose. For the statements marked EXACT or COLLECTIVE there, correctness does not rest on the reliability of any language model.

What does rest on human and machine judgement, and is stated here so that it can be checked independently, is the choice of definitions, the claim of novelty, the attribution of prior work, and the PROSE rows of Table 1.

**Draft note — not part of the submission.** Items marked <sup>†</sup> in the list below have *not* been consulted in the original at the time of writing; the bibliographic data were taken from secondary sources and independently cross-checked against Crossref, but the contents were not read. They are: [FT] Friedman–Tamari 1967, cited only for the origin of the lattice and of the term *loi demi-associative* (the phrase occurs in the title itself); [Gry] — *now consulted in the original* (Lith. Math. J. **62**(2), 192–206, 15 pages): it compares right-associated power towers with varying entries and heights and does not mention bracketings, binary trees, Catalan structures, rotations or any lattice, which confirms the description given in Section 8; [Tamari] Tamari 1962, cited only for the enumeration of bracketings and for the two combs being the extreme elements; and [Ste] Stembridge 1989, on Brunson’s order, cited only to place the question among its peer-reviewed neighbours. [Bru], [Gri] and [GW] were also identified late, in response to a referee report, but have since been consulted in the original and are no longer marked: all three fix the bracketing to the right comb and vary the entries, as now described in Section 8. None of the results of the present paper depends on these three items: they are cited for attribution and context, and every mathematical statement used in the proofs is either proved here or formalised in the accompanying Lean development. Before submission each of the three must be checked in the original, and this note removed by setting `\draftfalse`.

## References

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- [12] <sup>†</sup> D. Tamari, *The algebra of bracketings and their enumeration*, Nieuw Archief voor Wiskunde (3) **10** (1962), 131–146.
- [13] N. Zeilberger, *A sequent calculus for a semi-associative law*, Logical Methods in Computer Science **15** (2019), no. 1, 9:1–9:23.